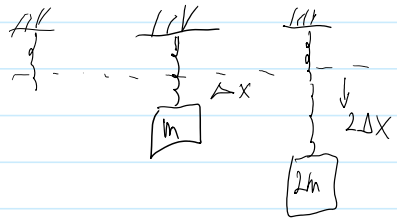
Lecture 2 - 29/04

Tuesday, May 12, 2026 7:59 PM



$F = -kx$

$F = ma = m\dot{v} = m\ddot{x}$

remider $\dot{x} = \frac{dx}{dt}$, $\ddot{x} = \frac{d^2x}{dt}$

$m\ddot{x} = -kx$

this is not a dynamical system because there is second deriv

a dynamical system is $\dot{\vec{x}} = F(x)$

$$\begin{cases} \dot{x} = v \\ m\dot{v} = -kx \end{cases}$$

$$\Downarrow$$

$$\begin{cases} \dot{x} = v \\ \dot{v} = -\frac{kx}{m} \end{cases}$$

this is a dynamical system

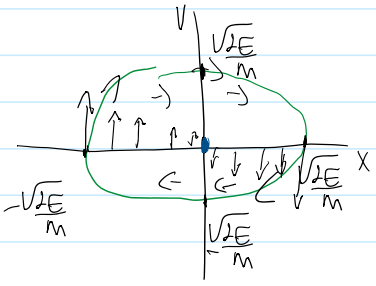
energy of the system

$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

let's check that energy is conserved ($\dot{E} = 0$)

$\dot{E} = mv\dot{v} + kx\dot{x} = mv\frac{-kx}{m} + kxv = -kxv + kxv = 0$

we can plot it



look at cases

1) if $E = 0 \Rightarrow v = 0$ and $x = 0$

since $\dot{E} = 0$, this won't change

$$\begin{cases} \dot{x} = v = 0 \\ \dot{v} = -\frac{k}{m}x = 0 \end{cases}$$

fixed point

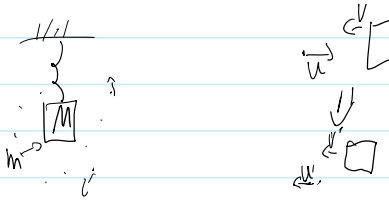
2) $E<0$, if we move E equation around, we get

$$\frac{v^2}{\left(\frac{2E}{m}\right)} + \frac{x^2}{\left(\frac{L^2}{k}\right)} = 1 \quad \leftarrow \text{an ellipse}$$

smaller E, smaller ellipse
we know the direction by looking at $\dot{x}$ and $\dot{v}$ at some points
the arrow map is the vector field
the trajectory are called orbits

notice $\vec{x}(t) = \vec{x}(t+T)$
smallest T is period

lets introduce friction (put it in air)



$F = ma = m\dot{v} = \dot{p}$

$$\begin{cases} mu + Mv = -mu' + Mv' \\ \frac{mu^2}{2} + \frac{Mv^2}{2} = \frac{mu'^2}{2} + \frac{Mv'^2}{2} \end{cases}$$

$$m(u+u') = M(v'-v)$$

div $\downarrow$ $m(u^2-u'^2) = M(v'^2-v'^2)$

$u-u' = v+v'$

$a^2-b^2 = (a-b)(a+b)$

$$u' = -v' - v + u$$

$$v' = \frac{(M+m)v - 2mu}{M+m} \qquad u' = \frac{2Mv + (M-m)u}{M+m}$$

We want to calc, what is big mass experiencing from many collisions

average

$$\langle \Delta p \rangle = M\langle (v'-v) \rangle = M\left\langle \frac{(M-m)v - 2mu}{M+m} - v \right\rangle = M\left\langle \frac{(M-m)v - 2mu - (M+m)v}{M+m} \right\rangle =$$

$$= M\left\langle \frac{-2mv - 2mu}{M+m} \right\rangle = M\left\langle \frac{-2m(v+u)}{M+m} \right\rangle \overset{222}{=} -\frac{2Mm}{M+m}(\langle v+u \rangle) = \frac{2Mm}{m+M}(\langle v \rangle + \langle u \rangle) =$$

$$= \frac{-2mM}{m+M} \cdot v$$

$\langle u \rangle$ is 0
since for every collision there an opposite one

$$F = \frac{\langle \Delta p \rangle}{\Delta t} \cdot \#\text{collisions} \sim -v$$

so now

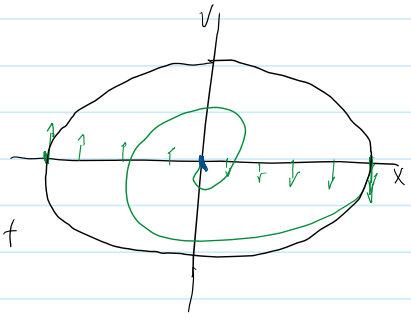
$F = -kx - b\dot{x}$

$m\ddot{x} = -kx - b\dot{x}$

$$\begin{cases} \dot{x} = v \\ \dot{v} = -\frac{k}{m}x - \frac{b}{m}v \end{cases}$$

$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

$\dot{E} = mv\dot{v} + kx\dot{x} = mv\left(-\frac{kx}{m} - \frac{bv}{m}\right) + kxv = -bv^2 \leq 0$

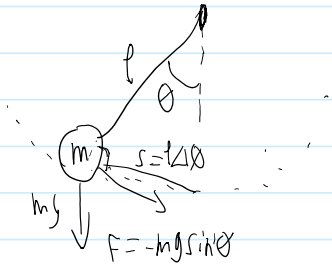


it will never reach the ellipse again
this trajectory is called Lapunov stable

Stable fixed point = trajectory will approach it

Fixed points
Periodic orbits
stable focus
unstable focus
Phase portraits
orbits
vector field
flow
$\dot{\vec{x}} = f(\vec{x})$

Pendulum



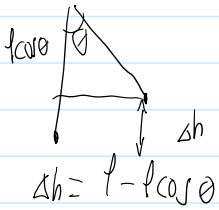
$\dot{x} = \lim_{\Delta t \to 0} l\frac{\Delta\theta}{\Delta x} = l\dot{\theta}$

$\ddot{x} = \frac{d\dot{x}}{dt} = \frac{d\,l\dot{\theta}}{dt} = l\frac{d\dot{\theta}}{dt} = l\ddot{\theta}$

$F = ma = m\ddot{x} = ml\ddot{\theta}$

$ml\ddot{\theta} = -mg\sin\theta$

$$\begin{cases} \dot\theta = w \\ \dot w = -\frac{g}{l}\sin\theta \end{cases}$$ this is non linear



$\Delta h = l - l\cos\theta$

$$E = \frac{mv^2}{2} + mgl(1-\cos\theta)$$

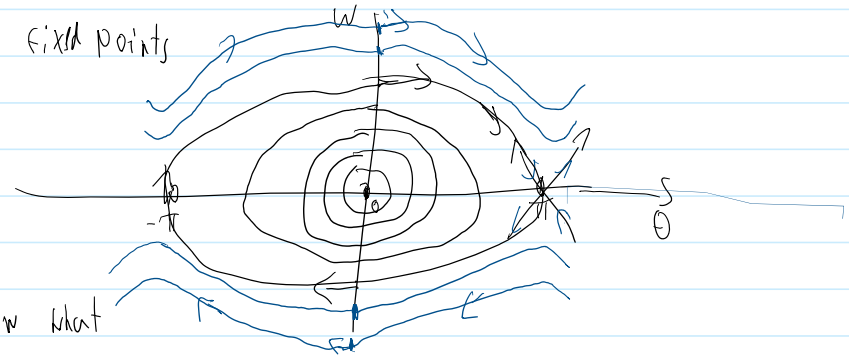
$$E = \frac{m(l\dot\theta)^2}{2} + mgl(1-\cos\theta)$$

$$E = \frac{m(lw)^2}{2} + mgl(1-\cos\theta)$$

$$\dot E = ml^2 w\dot w + mgl\sin\theta\cdot\dot\theta = ml^2 w\left(-\frac{g}{l}\sin\theta\right) + mgl\sin\theta\, w = 0$$

start by analyzing fixed points

$$\begin{cases} 0 = w \\ 0 = -\frac{g}{l}\sin\theta \end{cases}$$ we have ∞ fixed points



① $\theta \ll 1$ $\sin\theta \sim \theta$

$$\begin{cases} \dot\theta = w \\ \dot w = -\frac{g}{l}\theta \end{cases}$$

this is a liear approximation to show what happens locaLy to 0

② $|\theta + \pi| \ll 1$ change coordinates

$$\begin{cases} \theta = w \\ \dot w = \frac{g}{l}\theta \end{cases}$$

$$\begin{cases} x = \frac{\theta + w}{2} \\ y = \frac{\theta - w}{2} \end{cases}$$

$$\begin{cases} \theta = (x+y)\frac{l}{g} \\ w = x - y \end{cases}$$

$$\begin{cases} \dot x + \dot y = (x-y)\frac{g}{l} \\ \dot x - \dot y = \frac{g}{l}(x+y) \end{cases}$$

$$\dot x = 2\frac{g}{l}x$$

$$\dot y = -\frac{g}{l}y$$